

RIP_LitReview

by Sowrav Chowdhury

Submission date: 13-May-2022 11:15AM (UTC+0600)

Submission ID: 1835220071

File name: of_Respiratory_Inductance_Plethysmography_A_Narrative_Review.pdf (481.01K)

Word count: 4281

Character count: 23841

Healthcare Applications of Respiratory Inductance Plethysmography: A Narrative Review

Abstract. Accurate interpretation of respiratory functions permits early detection and treatment of life-threatening illnesses such as asthma, pneumonia, chronic obstructive pulmonary disease (COPD), and tuberculosis (TB), among others. With the advancement of technology, a diverse array of breath-monitoring devices has become available. The Breathing Inductance Plethysmography (RIP) is one of these devices that enables the acquisition of respiratory patterns and signals in a non-invasive manner. The purpose of this study is to provide an in-depth analysis of current breakthroughs in RIP devices, as well as to highlight significant applications and approaches in the modern day and to provide recommendations for future research areas. The article examined databases from IEEE, Science Direct, PubMed, and Google Scholar. We included publications that highlighted the critical enabling properties of healthcare applications. The research concluded that RIP is an effective and precise method for testing and monitoring respiratory function.

Keywords: Respiratory Inductance Plethysmography (RIP), COPD, wearable sensors, lung volume, non-invasive

1 Introduction

Breathing/respiratory function assessment led to early detection of probable respiratory disorders and provides valuable information for intervention. Numerous works on respiratory monitoring systems and pulmonary function assessment employing sensors have been conducted during the last few decades. Pulmonary function tests (PFT) are used to assess the vital signs and symptoms of patients with respiratory illnesses, whether they are suspected or recovering. The pneumotachometer (PnT) is the standard method in medical applications for assessing tidal volume (TD) change; however, using a PnT necessitates the individual to wear a mouthpiece, which indicates the change in internal ventilation. Spirometry and plethysmography with body box measurement are two more PFT procedures that take time and require participant cooperation [1], [2].

Substantial resources are empowered in recent years due SARS-CoV-2 pandemic to facilitate medical diagnosis, develop wearable sensors and create technology to identify numerous different pulmonary diseases. The use of wearable sensor technology in the health-monitoring arena has also increased significantly. Several investigations [3] on breath monitoring devices and pulmonary function testing with sensors have been conducted during the last few decades. Pulmonary function tests (PFT) are used to examine the health of individuals with respiratory illnesses, whether they are suspected or confirmed. They can help with diagnosis, therapy response monitoring, and

prevention and treatment recommendations. PFT and other respiration assessments such as chest wall motion, airflow and respiratory volumes, respiratory rate, resulting transcutaneous oxygen saturation (SpO₂), and respiratory CO₂ removal have all been monitored as a result of these efforts. The equipment and methods for assessing respiration are classified in Table 1.

1

Table 1. Categories of Sensing Principle for Respiratory Monitoring Devices and Methods

Measured Technique	Category	Measured Parameters	Sensor Position
RIP, EIP ¹ , Full Body Plethysmography, Spirometry	Lung Movement	ABD ² & THC ³ Circumference change, FEV ⁴ , FVC ⁵ , FRC ⁶ , TLC ⁷ , RV ⁸	ABD and chest wall
ABG ⁹ Test	Arterial gas concentration	pH, P _a O ₂ ¹⁰ , P _a CO ₂ ¹¹ , SBC _e ¹² , HCO ₃ ⁻ ¹³ , (tCO ₂ (P) _e) ¹⁴	Blood gas measurement Arterial gas concentration Peripheral organ or Nasal/oral area

Apart from the traditional techniques of respiratory parameter measurements, respiratory inductive plethysmography (RIP), is a noninvasive measurement technique of lung ventilation that consists of two elastic belts one in the abdomen around the umbilicus and one in the rib cage around the nipple line, to assess spontaneous tidal breathing by the movement of the chest during respiration. Generally, RIP can be done on subjects in around 5-10 mins depending on the condition and calibration [4]. Conventionally the RIP bands are incorporated with a transducer essentially with coils that are connected to an analog circuit which is a variable frequency LC oscillator. The coils act as an inductive element and therefore the frequency and amplitude can be calculated to find the inductance. The length of the belt varies with each breath,

¹ Electrical Impedance Plethysmography

² Abdomen

³ Thoracic

⁴ Forced Expiratory Volume in 1 second

⁵ Forced Vital Capacity

⁶ Functional Residual Capacity

⁷ Total Lung Capacity

⁸ Residual Volume

⁹ Arterial Blood Gas

¹⁰ Arterial O₂ Partial Pressure

¹¹ Arterial CO₂ Partial Pressure

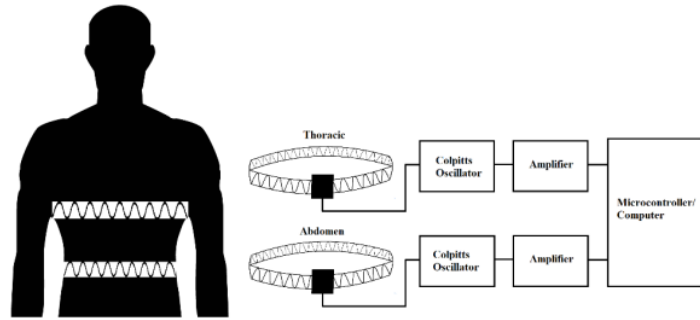
¹² Concentration of bicarbonate

¹³ Metabolic acidosis/alkalosis

¹⁴ Total CO₂

and the coils' self-inductance varies accordingly. By monitoring changes in the coils' self-inductance, differences in the circumference or cross-sectional area of the rib cage and belly can be observed. The fundamental function of RIP is depicted in Figure 1.

Fig. 1. Basic block diagram for RIP



While most of the existing technologies show promise, the major advantage of RIP over PMT is that the tidal breathing in patients who are unable to cooperate due to their old age or disease severity can also be examined through an easy set of equipment [5]. On top of that, the RIP system can also be implemented for individuals in an ambulatory environment [6,7]. The ease of this system, the device, and the method are becoming of big interest to researchers and clinicians. Rather than just a comprehensive analysis of RIP devices, this work intends to inform the reader on the fundamental enabling aspects of RIP devices, their prospective applications in the health sector via illness detection, and design highlights in a variety of settings and applications.

2 Key Enabling Features

2.1 Advancement of RIP

Over time, the RIP device evolved depending on the applications. One important aspect of the RIP is that it is oscillator-based hardware that requires calibration over some time. Additionally, it also requires several inductive sensors to obtain mutual inductance between RIP sensors. To address this issue with RIP, Cohen et al [8]; Zhang et al. [9] have suggested that the frequencies of the thoracic and abdominal band oscillators must be suitably spaced in an attempt to lessen modulation and avoid frequency locking. Even though a respiration monitoring system is typically equipped with multiple inductive sensors, establishing a stable oscillator for each sensor while maintaining frequency separation is challenging. For a battery-driven portable device,

energy consumption is critical and hence a small printed circuit board size becomes essential for mobility. Therefore, if multiple inductive sensors are involved, a traditional RIP design employing an LC oscillator increases chip count and power management intricacy. To mitigate this problem, Sinton [10] devised a RIP system that uses a traditional Electrical Impedance Plethysmograph (EIP) to quantify the coil's self-inductance, since a steady current travels through the belts, this approach does not require an LC oscillator. Sinton's method incorporates a continuous AC signal which resolves the frequency locking but requires high power which makes the system impossible to be portable. Zhang et al. [10] enact a Pulse Amplitude Modulation (PAM) technique for indirect inductance measurement without the need for a variable-frequency oscillator [8, 10, 12,]. During the pulse excitation phase, a strong current from a constant-current source is momentarily and routinely given to the RIP sensor, and its inductance is measured. Unlike Sinton's method, the feeding current does not pass through the inductive sensor continually. As a result of the large feeding current at the time, the PAM method significantly decreases power usage while preserving sensitivity. Contemporary developments have completely diminished the use of an LC oscillatory circuit by using a high-performance, low-power embedded microprocessor that instantly digitizes the inductance, therefore, requiring no calibration and allowing the storage of data locally and simultaneously transmitting it wirelessly over computer and cloud using *PneuRIP*TM [13]. As a result, it is recommended that the user account for advancements in the RIP device prior to applying it to the intended function.

2.2 Wearability and Mobility

Continuous monitoring of respiration is crucial for detecting and anticipating high-risk situations where an appropriate monitoring technique may pave the way to life-saving conditions. The implementation of RIP in a wearable form factor allows to monitor PFT and breathing pattern (BP) of the user in a completely unrestrained environment, therefore allowing to measure data while the individual can carry on their daily activities. The mobility of RIP is also advantageous in ambulatory situations, where PFT and BP are critical to measuring. Having such a system embedded in an ambulance might help diagnose patients, and provide critical reports for medics.

Wearable shirts with embedded sensors for monitoring vital indicators, including blood pressure, are available from Carre Technologies in Canada. The Hexoskin Smart Shirt (HX) was created with activity sensors in mind for usage in athletics and therapeutic settings [14, 15]. This sort of wearable shirt can also be highly beneficial in other disciplines of research, such as cigarette smoking monitoring [16].

3 Applications of RIP Sensor in Disease Recognition

From a clinical standpoint, 'respiratory activity' provides aspects such as respiratory rate and depth, as well as quantitative information on the degree of gas exchange that is taking place. Continuous respiratory activity monitoring is crucial for recognizing or anticipating high-risk diseases, and correct monitoring techniques may save lives.

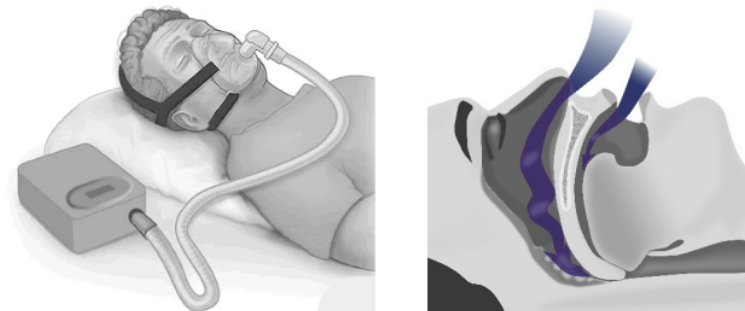
3.1 Apnea and Hypopnea

Obstructive sleep apnea-hypopnea syndrome (OSAHS) is a serious medical disorder that has become more common in recent decades. It is a major source of illness and death in both developed and developing countries [17].

For the identification of respiratory events during sleep studies, the American Academy of Sleep Medicine (AASM) advises using an oronasal thermistor and a nasal cannula [18]; nevertheless, these sensors are unpleasant for some individuals and can even disrupt their sleep. They can be inaccurate since patients can move them around or even remove them during sleep examinations. Fig. 2(a) depicts continuous positive airway pressure (CPAP) a method frequently used treatment for obstructive sleep apnea. Sabil, A et al. [19] suggested that the combination of tracheal sounds and RIP could be used as a reliable method for the detection and characterization of respiratory events without the need to place sensors on the face of patients and could be used either as an alternative or in combination with the recommended American Academy of Sleep Medicine sensors.

When given to young babies, anesthesia increases the risk of apnea in the postoperative period, a condition known as postoperative apnea (POA) [20, 21]. There have been reports of both central and obstructive apnea [22]. The absence of inspiratory effort characterizes central apneas, due to the obstruction in the pharyngeal airway, obstructive apneas induce respiratory effort but no inspiratory outflow. Both central and obstructive apnea is life-threatening since the majority of newborns require medical intervention to reestablish breathing; fatalities have been documented [23]. Fig 2. (b) models the airway blocking due to apnea.

Fig. 2. (a) On the left shows a patient in sleep while using a CPAP (b) On the right patient with OSA with a tissue blocking the airway



5 An Automated Unsupervised Respiratory Event Analysis (AUREA) to identify breathing patterns acquired with dual belt RIP was implemented by [24]. The results given in this paper show that AUREA may be used to swiftly categorize breathing patterns in signals obtained from RIP, sample by sample. The AUREA pattern classifications have also been used to evaluate preterm neonates' breathing patterns after the endotracheal tube (ETT) has been withdrawn. To construct a predictor of extubation readiness in the extreme neonatal newborn, both AUREA patterns and AUREA metrics will be employed. As a result, analyzing the patterns and metrics produced from AUREA may indicate breathing trends and predict sensitive infants with the potential for POA.

3.2 Neuromuscular Diseases

17 Neuromuscular diseases are a wide term that refers to a variety of disorders that affect muscle function, either directly via pathologies of the voluntary muscle or indirectly through pathologies of the peripheral nervous system or neuromuscular junctions. Respiratory failure in neuromuscular illness might manifest as an abrupt occurrence or a chronic, progressive disease [3]. Approaches to measuring breathing in neuromuscular illnesses can be costly, time-consuming, and stressful for patients. Better assessment procedures, such as in-home examinations, are being explored. The *PneuRIP* might be a more efficient way to assess a patient's lung health. In [13] an outpatient clinic environment was set up and evaluates the efficacy of the *PneuRIP* in adolescents with neuromuscular illness and evaluated their RIP parameters to those of age-matched healthy individuals. The *PneuRIP* technique successfully evaluated the Work of Breathing (WOB) indices and respiratory patterns and has proven to be effective in recognizing abnormalities in WOB indices and breathing patterns in patients with the neuromuscular illness.

3.3 Other Restrictive and Obstructive Lung Diseases

In medical practice, body plethysmography can deduce both restrictive and obstructive lung diseases such as Asbestosis, Pulmonary fibrosis, Sarcoidosis, Scoliosis, Asthma, Bronchiectasis, and Emphysema Cystic fibrosis, and Chronic bronchitis. Along with these, other cardiovascular diseases can also be monitored from the PFT using RIP. Therefore, the RIPs can be calibrated and/or designed for the identification of a specific disease in a non-invasive manner and in a varied set of ambulatory environments. Table 2 lists important research works with RIP.

Table 2. Key Findings of other research papers along with disease recognition and device used

Sl No.	Paper Title	Published Year	Device	Disease Recognition
1	Evaluation of RIP Sensor Calibration Stability for Daily Estimation of Lung Volume [6]	2017	SleepSense Inductive Plethysmography, S.L.P. Inc.	Monitoring of Cigarette Smoking
2	Respiratory Monitoring by Inductive Plethysmography in Unrestrained Subjects Using Position Sensor-Adjusted Calibration [27]	2010	LifeShirt TM VivoMetrics, Ventura	Calibration Technique for different disease recognition
3	Automatic unsupervised respiratory analysis of infant respiratory inductance plethysmography signals [23]	2020	AUREA	Postoperative Apnea detection in infants
4	The design of RIP belts impacts the reliability and quality of the measured respiratory signals [28]	2021	-	Sleep Apnea Detection
5	Vision-Based Heart and Respiratory Rate Monitoring During Sleep – A Validation Study for the Population at Risk of Sleep Apnea [30]	2019	Point Grey Firefly MV	Non-contact monitoring of respiratory rate
6	Effect of pediatric ventilation wearing technique on work of breathing [29]	2022	Viasys, Healthcare, Respiband Plus	Patient effort between a pressure regulating intermittent mode of ventilation with or without pressure support (PS)
7	Enhanced Breathing Pattern Detection during Running Using Wearable Sensors [29]	2021	Hexoskin TM	Developed a wearable garment for RIP signals

4 Design and Performance of RIP

Over the years, RIP's design and performance have been rigorously tested. Due to the fact that the RIP is built on belts, the design of the wire woven or sewn along the belt, as well as its specifications, have an effect on the sensor characteristics of the RIP. The study in [26] examined the effect of the sine wave pattern's step size on the properties of RIP sensors by fabricating three RIP sensors with the same wire pattern (sine wave), height (2.2 cm), and length (30 cm), but with varying step sizes, as seen in Fig. 3. Each of these belts' inductance was determined using an Agilent 4285A Precision LCR Meter. Fig. 3 illustrates the sensitivity and linearity of sensors with varying step sizes. As seen in Fig. 3, the smaller the step size, the greater the sensitivity. There is a tiny variance in linearity between the 1 cm and 1.5 cm step sizes. However, linearity

significantly reduces with a step size of 3 cm. It is thus prudent to keep the step size modest in order to maintain linearity.

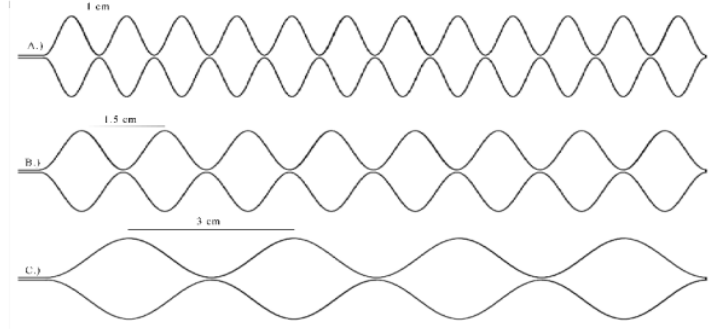


Fig. 3. RIP belt coil design with three different sine wave patterns with step sizes of A.) 1 cm, B.) 1.5cm C.) 3cm [26]

The researchers also contributed to investigating the placement of RIP sensors. The authors in [26] scrutinized three different attachment options for the RIP, (i) RIP belt secured using tension only; (ii) RIP belt held in place by adhesive and tension; (iii) RIP belt sewn into a T-Shirt. From the findings over 24 hours, the method with the RIP belt stitched inside a T-Shirt had the highest performance. This approach has been proven to be more resistant to abnormalities induced by body movement. The RIP was only to be removed during showering, and participants in the trial were instructed to go about their everyday routines. The calibration coefficients of the system based merely on the belt or belt+adhesive become unreliable over time. This happened as a result of their day-to-day activities, making them unreliable and inconvenient. Table 2 demonstrates the system's average overall volume and standard deviation, whilst Table 3 reveals the data as calculated Root Mean Square Error (RMSE).

Table 2. System Comparison in terms of Volume in Liters (L) [13]

System	Average Volume (L)		
	Subject 1	Subject 2	Subject 3
Belt	0.577±0.093	0.999±0.212	1.094±0.295
Belt+Adhesive	0.761±0.445	1.295±0.272	0.994±0.517
Belt+Shirt	0.831±0.111	0.735±0.116	0.798±0.127

Table 3. System Comparison in terms of RMS in Liters (L) [13]

System	Average Volume (RMS)		
	Subject 1	Subject 2	Subject 3
Belt	0.223	0.199	0.294
Belt+Adhesive	0.039	0.495	0.194
Belt+Shirt	0.031	0.066	0.002

In [13], the researchers have conducted a Benchtop testing on the *PneuRIPTM* to determine the circuitry and instrument design, reliability, responsiveness, and repeatability. The RIP bands were mounted on a laboratory-grade rocker/waver platform to mimic different breathing situations, the tilt angles and speed were changed. Over two days, the testing ran without taking any pauses. The benchtop testing was performed along with the RespiTrace system (Sensormedics) a RIP system from 1991 [31]. As typical breathing frequencies and phase angles, both systems *PneuRIPTM* and RespiTrace produce similar results but the phase angle variance was substantially reduced using *PneuRIPTM* for low and high phase angles, as well as low and high breathing frequency. *PneuRIPTM* is more precise and reliable at greater and lower phase angles because the mechanical fluctuation is less. Children with respiratory distress frequently have large phase angles, which *PneuRIPTM* is intended to detect. A study [26] suggested that the step size should be smaller to achieve good linearity. Regarding the placement of the device, it was discouraged to use the RIP device with a belt and adhesive.

5 Discussion & Conclusion

Respiratory Inductance Plethysmography (RIP) has been shown to be an effective and safe tool for measuring and monitoring human respiratory function. According to the study reviewed, RIP has shown to be a very helpful assessment tool for examining the respiratory system's physiology in a range of circumstances, both pathological and non-pathological, allowing health practitioners to diagnose sickness at an early stage.

It was determined that the creation and improvement of the RIP device occurred over time. With advancements in technology, the calibration process and worries about mutual inductance amongst RIP sensors have been improved. In terms of mobility and wearability, it was obvious that the RIP sensor had been integrated into the Smart Shirt, which makes it quite useful in ambulatory situations. RIP is commonly utilized in healthcare applications. The device was very effective in detecting obstructive sleep apnea-hypopnea syndrome. Additionally, the RIP device is beneficial for gathering data from neonates and young patients. Additionally, the RIP gadget might be used to identify and monitor neuromuscular abnormalities. The PneuRIP was shown to be very effective in adolescents with neuromuscular illness. According to the literature review, the design and placement of RIP had a major impact on the system's performance and durability.

To our knowledge, this is the only review paper that provides a concise summary of the medical field's usage of RIP devices. The review, however, might be significantly strengthened by integrating a study of the RIP device's accuracy in a wider population and in senior patients. Future development on the RIP device's design has the potential in key respiratory disease recognition.

References

1. Rosenfeld M, Allen J, Arets B et al. (2013) An Official American Thoracic Society Workshop Report: Optimal Lung Function Tests for Monitoring Cystic Fibrosis, Bronchopulmonary Dysplasia, and Recurrent Wheezing in Children Less Than 6 Years of Age. *Annals of the American Thoracic Society* 10:S1-S11. doi: 10.1513/annalsats.201301-017st
2. Mayer O, Clayton R, Jawad A et al. (2003) Respiratory Inductance Plethysmography in Healthy 3- to 5-Year-Old Children *. *Chest* 124:1812-1819. doi: 10.1378/chest.124.5.1812
3. Yuki K, Fujiogi M, Koutsogiannaki S (2020) COVID-19 pathophysiology: A review. *Clinical Immunology* 215:108427. doi: 10.1016/j.clim.2020.108427
4. Strang A, Ryan L, Rahman T et al. (2018) Measures of respiratory inductance plethysmography (RIP) in children with neuromuscular disease. *Pediatric Pulmonology* 53:1260-1268. doi: 10.1002/ppul.24134
5. Beydon N, Davis S, Lombardi E et al. (2007) An Official American Thoracic Society/European Respiratory Society Statement: Pulmonary Function Testing in Preschool Children. *American Journal of Respiratory and Critical Care Medicine* 175:1304-1345. doi: 10.1164/rccm.200605-642st
6. Hutchison A, Leclerc F, Nève V et al. (2014) The Respiratory System. *Pediatric and Neonatal Mechanical Ventilation* 55-112. doi: 10.1007/978-3-642-01219-8_4
7. Ramos-Garcia R, Imtiaz M, Sazonov E, Tiffany S (2017) Evaluation of RIP sensor calibration stability for daily estimation of lung volume. 2017 Eleventh International Conference on Sensing Technology (ICST). doi: 10.1109/icsenst.2017.8304419
8. Cohen K, Panescu D, Booske J et al. (1994) Design of an inductive plethysmograph for ventilation measurement. *Physiological Measurement* 15:217-229. doi: 10.1088/0967-3334/15/2/009
9. Zhang Z, Zheng J, Wu H et al. (2012) Development of a Respiratory Inductive Plethysmography Module Supporting Multiple Sensors for Wearable Systems. *Sensors* 12:13167-13184. doi: 10.3390/s121013167
10. Sinton A, Suntheralingam R (1988) Respiratory inductance plethysmography with an electrical impedance plethysmograph. *Medical & Biological Engineering & Computing* 26:213-217. doi: 10.1007/bf02442267

11. Zhang Z, Yu M, Wu T, Wu J (2006) Design of a Wearable Respiratory Inductive Plethysmograph and Its Applications. *Space Medicine & Medical Engineering*
12. Zheng J, Zhang Z, Wu T, Zhang Y (2007) A wearable mobihealth care system supporting real-time diagnosis and alarm. *Medical & Biological Engineering & Computing* 45:877-885. doi: 10.1007/s11517-007-0221-y
13. Rahman T, Page R, Page C et al. (2017) pneuRIPTM: A Novel Respiratory Inductance Plethysmography Monitor. *Journal of Medical Devices*. doi: 10.1115/1.4035546
14. Phillips, M. B., Beach, J., Cathey, M., Lockert, J., & Satterfield, W. (2017). Reliability and Validity of the Hexoskin Wearable Body Metrics Telemetry Shirt. *Journal of Sport and Human Performance*, 5(2). doi: 10.12922/jshp.v5i2.108
15. Cherif, N. H., Neila Mezghani, Nathaly Gaudreault, Youssef Ouakrim, I. Mouzoune and Pierre Boulay. "Physiological Data Validation of the Hexoskin Smart Textile." *BIODEVICES* (2018).
16. Lopez-Meyer P, Tiffany S, Patil Y, Sazonov E (2013) Monitoring of Cigarette Smoking Using Wearable Sensors and Support Vector Machines. *IEEE Transactions on Biomedical Engineering* 60:1867-1872. doi: 10.1109/tbme.2013.2243729
17. Sale S (2010) Neonatal apnoea. *Best Practice & Research Clinical Anaesthesiology* 24:323-336. doi: 10.1016/j.bpa.2010.04.002
18. Mansukhani M, Kolla B, Ramar K (2014) International Classification of Sleep Disorders 2 and American Academy of Sleep Medicine Practice Parameters for Central Sleep Apnea. *Sleep Medicine Clinics* 9:1-11. doi: 10.1016/j.jsmc.2013.10.006
19. Sabil A, Marien C, LeVaillant M et al. (2020) Diagnosis of sleep apnea without sensors on the patient's face. *Journal of Clinical Sleep Medicine* 16:1161-1169. doi: 10.5664/jcsm.8460
20. Brown K, Aoude A, Galiana H, Kearney R (2008) Automated respiratory inductive plethysmography to evaluate breathing in infants at risk for postoperative apnea. *Canadian Journal of Anesthesia/Journal canadien d'anesthésie* 55:739-747. doi: 10.1007/bf03016346
21. Kurth C, LeBard S (1991) Association of Postoperative Apnea, Airway Obstruction, and Hypoxemia in Former Premature Infants. *Anesthesiology* 75:22-26. doi: 10.1097/0000542-199107000-00005

22. Malviya S, Swartx J, Lerman J (1993) Are All Preterm Infants Younger than 60 Weeks Postconceptual Age at Risk for Postanesthetic Apnea?. *Anesthesiology* 78:1076-1081. doi: 10.1097/0000542-199306000-00009
23. Davidson A, Morton N, Arnup S et al. (2015) Apnea after Awake Regional and General Anesthesia in Infants. *Anesthesiology* 123:38-54. doi: 10.1097/aln.0000000000000709
24. Robles-Rubio C, Kearney R, Bertolizio G, Brown K (2020) Automatic unsupervised respiratory analysis of infant respiratory inductance plethysmography signals. *PLOS ONE* 15:e0238402. doi: 10.1371/journal.pone.0238402
25. Landon C (2009) Novel Methods of Ambulatory Physiologic Monitoring in Patients With Neuromuscular Disease. *Pediatrics* 123:S250-S252. doi: 10.1542/peds.2008-29521
26. Ngo H, Nguyen C, Nguyen T, Van Vo T (2013) A Portable Respiratory Monitor Using Respiratory Inductive Plethysmography. *IFMBE Proceedings* 222-225. doi: 10.1007/978-3-642-32183-2_57
27. Brüllmann G, Fritsch K, Thurnheer R, Bloch K (2010) Respiratory Monitoring by Inductive Plethysmography in Unrestrained Subjects Using Position Sensor-Adjusted Calibration. *Respiration* 79:112-120. doi: 10.1159/000212117
28. Montazeri K, Jonsson S, Agustsson J et al. (2021) The design of RIP belts impacts the reliability and quality of the measured respiratory signals. *Sleep and Breathing* 25:1535-1541. doi: 10.1007/s11325-020-02268-x
29. Harbour E, Lasshofer M, Genitrini M, Schwameder H (2021) Enhanced Breathing Pattern Detection during Running Using Wearable Sensors. *Sensors* 21:5606. doi: 10.3390/s21165606
30. Zhu K, Li M, Akbarian S et al. (2019) Vision-Based Heart and Respiratory Rate Monitoring During Sleep – A Validation Study for the Population at Risk of Sleep Apnea. *IEEE Journal of Translational Engineering in Health and Medicine* 7:1-8. doi: 10.1109/jtehm.2019.2946147
31. Bar-Yishay E, Putilov A, Einav S (2003) Automated, real-time calibration of the respiratory inductance plethysmograph and its application in newborn infants. *Physiological Measurement* 24:149-163. doi: 10.1088/0967-3334/24/1/311

ORIGINALITY REPORT

29%

SIMILARITY INDEX

19%

INTERNET SOURCES

26%

PUBLICATIONS

15%

STUDENT PAPERS

PRIMARY SOURCES

1

H. T. Ngo, C. V. Nguyen, T. M. H. Nguyen, Toi Van Vo. "Chapter 57 A Portable Respiratory Monitor Using Respiratory Inductive Plethysmography", Springer Science and Business Media LLC, 2013

Publication

2%

2

Zhengbo Zhang, Jiewen Zheng, Hao Wu, Weidong Wang, Buqing Wang, Hongyun Liu. "Development of a Respiratory Inductive Plethysmography Module Supporting Multiple Sensors for Wearable Systems", Sensors, 2012

Publication

2%

3

Abigail Strang, Lauren Ryan, Tariq Rahman, Sona Balasubramanian, Jobayer Hossain, Robert Heinle, Thomas H. Shaffer. "Measures of respiratory inductance plethysmography (RIP) in children with neuromuscular disease", Pediatric Pulmonology, 2018

Publication

1%

4

Submitted to Avondale College of Higher Education

Student Paper

1%

5	journals.plos.org Internet Source	1 %
6	www.mdpi.com Internet Source	1 %
7	Submitted to Swinburne University of Technology Student Paper	1 %
8	link.springer.com Internet Source	1 %
9	Carlos A. Robles-Rubio, Robert E. Kearney, Gianluca Bertolizio, Karen A. Brown. "Automatic unsupervised respiratory analysis of infant respiratory inductance plethysmography signals", PLOS ONE, 2020 Publication	1 %
10	Submitted to Polk State College Student Paper	1 %
11	Submitted to San Joaquin Valley College Student Paper	1 %
12	www.x-mol.com Internet Source	1 %
13	Anand Kumar Srivastava, Yugal Kumar, Pradeep Kumar Singh. "A Rule-Based Monitoring System for Accurate Prediction of Diabetes", International Journal of E-Health and Medical Communications, 2020	1 %

14	Submitted to University of Wales Swansea Student Paper	1 %
15	Submitted to University of Western Sydney Student Paper	1 %
16	Submitted to Florida Gulf Coast University Student Paper	1 %
17	Submitted to Jawaharlal Nehru University (JNU) Student Paper	1 %
18	Submitted to Idaho State University Student Paper	1 %
19	Submitted to Rhodes University Student Paper	1 %
20	Cote, C.J.. "1 Pre-operative preparation for anaesthesia and surgery", Bailliere's Clinical Anaesthesiology, 199612 Publication	1 %
21	www.frontiersin.org Internet Source	1 %
22	Submitted to Ivy Tech Community College Central Office Student Paper	1 %
23	J. Kyle Bohannon, Anna N. Honko, Rebecca J. Reeder, Kurt Cooper et al. "Comparison of	<1 %

respiratory inductive plethysmography versus head-out plethysmography for anesthetized nonhuman primates in an animal biosafety level 4 facility", Inhalation Toxicology, 2016

Publication

24

Obaid U. Khurram, Heather M. Gransee, Gary C. Sieck, Carlos B. Mantilla. "Automated evaluation of respiratory signals to provide insight into respiratory drive", Respiratory Physiology & Neurobiology, 2022

Publication

<1 %

25

www.distrofiamuscular.net

Internet Source

<1 %

26

dblp.dagstuhl.de

Internet Source

<1 %

27

health.embs.org

Internet Source

<1 %

28

Submitted to Asia Pacific University College of Technology and Innovation (UCTI)

Student Paper

<1 %

29

Submitted to Savitribai Phule Pune University

Student Paper

<1 %

30

Williams, R.K.. "Spinal anesthesia in infants", Techniques in Regional Anesthesia and Pain Management, 199907

Publication

<1 %

31	Wwww.rch.org.au Internet Source	<1 %
32	"Abstracts", Journal of Sleep Research, 2020 Publication	<1 %
33	Javier Gracia-Tabuenca, Ville-Pekka Seppä, Milla Jauhiainen, Marita Paassilta, Jari Viik, Jussi Karjalainen. "Tidal breathing flow profiles during sleep in wheezing children measured by impedance pneumography", Respiratory Physiology & Neurobiology, 2020 Publication	<1 %
34	Submitted to University of Minnesota System Student Paper	<1 %
35	Submitted to Cranfield University Student Paper	<1 %
36	Haythem Rehouma, Rita Noumeir, Sandrine Essouri, Philippe Jouvét. "Advancements in Methods and Camera-Based Sensors for the Quantification of Respiration", Sensors, 2020 Publication	<1 %
37	Submitted to Heriot-Watt University Student Paper	<1 %
38	eprints.gla.ac.uk Internet Source	<1 %
39	www.ncbi.nlm.nih.gov Internet Source	<1 %

40	www.tandfonline.com Internet Source	<1 %
41	ieeexplore.ieee.org Internet Source	<1 %
42	www.fbscience.com Internet Source	<1 %
43	"Cotes' Lung Function", Wiley, 2020 Publication	<1 %
44	Www.dailymail.co.uk Internet Source	<1 %
45	docksci.com Internet Source	<1 %
46	Peter J F M Merkus. "Setting the stage for CFTR modulator studies in infants", Thorax, 2014 Publication	<1 %
47	www.journalsleep.org Internet Source	<1 %
48	Submitted to Bond University Student Paper	<1 %
49	Sophia Schiza, Anita Simonds, Winfried Randerath, Francesco Fanfulla et al. "Sleep laboratories reopening and COVID-19: a European perspective", European Respiratory Journal, 2021	<1 %

50	iopscience.iop.org Internet Source	<1 %
51	"Pediatric and Neonatal Mechanical Ventilation", Springer Science and Business Media LLC, 2015 Publication	<1 %
52	B Vogt, C Falkenberg, N Weiler, I Frerichs. "Pulmonary function testing in children and infants", Physiological Measurement, 2014 Publication	<1 %
53	Ibrahim Sadek, Bessam Abdulrazak. "Contactless remote monitoring of sleep: evaluating the feasibility of an under-mattress sensor mat in a real-life deployment", Health Systems, 2022 Publication	<1 %
54	Justin Whitlock, Joshua Sill, Shubham Jain. "A-spiro: Towards continuous respiration monitoring", Smart Health, 2020 Publication	<1 %
55	dokumen.pub Internet Source	<1 %

RIP_LitReview

GRADEMARK REPORT

FINAL GRADE

/0

GENERAL COMMENTS

Instructor

PAGE 1

PAGE 2

PAGE 3

PAGE 4

PAGE 5

PAGE 6

PAGE 7

PAGE 8

PAGE 9

PAGE 10

PAGE 11

PAGE 12